

Methodology Note: Container Port Performance Index (CPPI)

June 10, 2026

1. Overview

The Container Port Performance Index (CPPI) is produced jointly by the World Bank and S&P Global Market Intelligence. It provides an objective, quantitative benchmark of container port efficiency based on vessel time in port. This note describes the data sources, construction of the two component indices, their combination into the final CPPI, and the approach used to produce a comparable multi-year time series.

The CPPI measures performance at the port level, not at the level of individual container terminals. In cases where a port comprises multiple container terminals, potentially operated by different companies and with different levels of productivity, the CPPI reflects the aggregate performance observed across the port calls included for that port. Accordingly, the index should be interpreted as a port-level benchmark and not as an assessment of any single terminal within a port.

2. Data Sources

The CPPI relies on a unique operational dataset provided by major shipping lines through S&P Global Market Intelligence Port Performance Program, which is integrated with AIS-derived vessel movement records.

2.1 Shipping line participation

The S&P Global Market Intelligence Port Performance Program started in 2009. It includes 10 of the largest global liner shipping companies, representing approximately 75–80 percent of the global container fleet capacity. The liner shipping companies provide the program with a series of data points, including operational time stamps and other information, such as move counts for each port call undertaken globally. These carriers provide monthly time-stamped data for container vessel port calls, covering their entire operational networks and subsidiaries.

2.2 Data transmission and mapping

Shipping lines transmit data directly to S&P Global Market Intelligence Port Performance Program, which then performs validation and standardization, verifying call data with AIS tracks using geofences. These are digital perimeters set around relevant port facilities and

operating areas to determine exactly when and where a vessel enters, berths, or departs. Only verified port-call records (with matching AIS and timeliness data) are included, with a reported 95 percent match rate.

3. Core Metric: Vessel Time in Port

The CPPI relies on detailed port call-level data for container vessels, derived from AIS signals matched with structured vessel and port call information. Each port call includes timestamps for six key events:

- Key event 1: arrival at anchorage or pilot station,
- Key event 2: ship movement to berthing place,
- Key event 3: start of cargo operations,
- Key event 4: end of cargo operations,
- Key event 5: departure from berth,
- Key event 6: exit from port limits.

Each port call also includes vessel characteristics (TEU capacity) and data on the call size, number of cranes deployed, and other relevant details. The data undergo rigorous cleaning to remove incomplete, inconsistent, or duplicated records.

The time between events 5 and 6 — the period after the ship leaves berth until it exits port limits — is excluded from CPPI calculations. This is because any port performance loss that pertains to departure delays, such as pilot or tug availability, readiness of the mooring gang, channel access and water depths, forecasting completion time, communication, and ship readiness, will be incurred while the ship is still alongside the berth and will already be included in the CPPI. Operations carried out in a port, but after departing from a berth, such as bunkering, repairs, or simply waiting in a safe area, are excluded from the CPPI, as they are not influenced by the operational performance of the port.

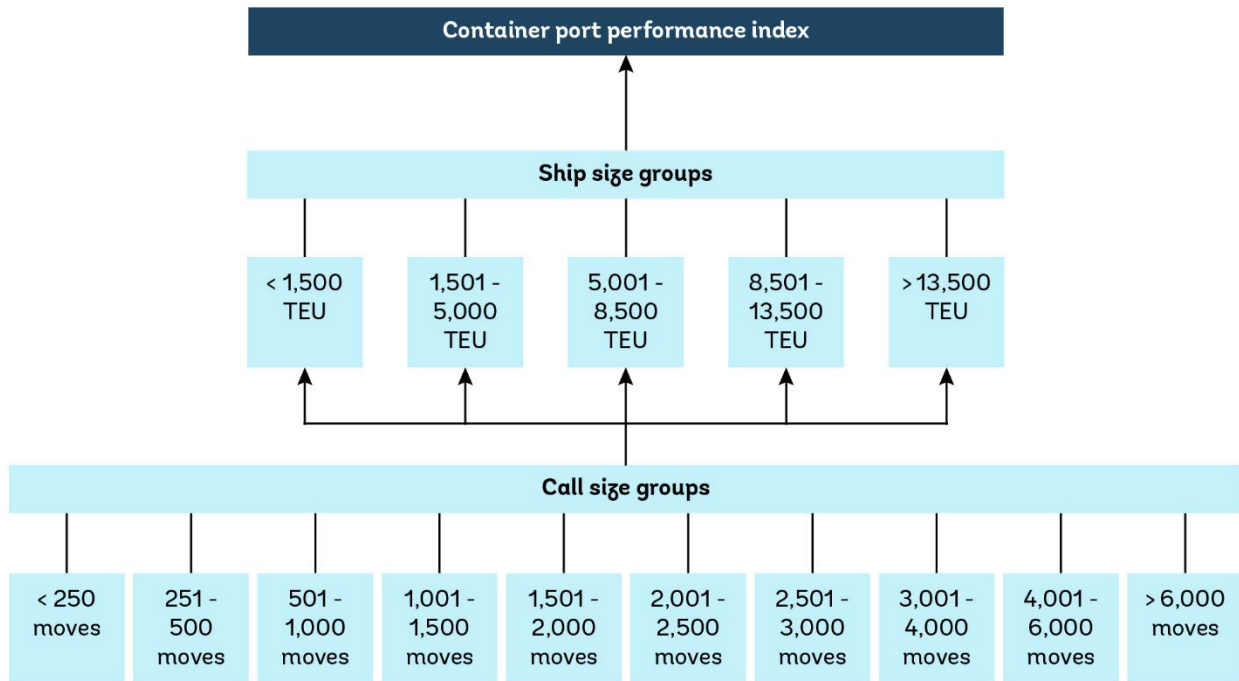
4. Calculating the CPPI index

The CPPI is constructed from two methodologically distinct approaches, each capturing a different dimension of port performance. The two are computed independently and then combined. Before either index is computed, the underlying port call data is grouped and cleaned to ensure that performance comparisons are made on a like-for-like basis. Port calls are stratified by ship size category and call size band, and observations that fall outside acceptable thresholds are excluded. With this cleaned, stratified dataset as a common input, two methodologically distinct approaches are then applied independently to produce the component indices: the administrative approach, and the statistical approach. The two component indices are then averaged into the overall CPPI index.

4.1 Grouping and cleaning the CPPI dataset

The structure of the CPPI is shown in Figure 1. To ensure comparability across ports of different scales, data is grouped into five ship size categories: feeders: <1,500 TEUs, intra-regional: 1,501 TEUs–5,000 TEUs, intermediate: 5,001 TEUs–8,500 TEUs, neo-panamax: 8,501 TEUs–13,500 TEUs, and ultra-large container carriers: >13,500 TEUs. The five ship size groups were based on where ships might be deployed and the similarities of ships within each group. For each category, there are ten different bands for call size. The ten call size groups were selected to ensure a similar level of crane intensity within each group. This prevents smaller feeder ports from being benchmarked against major global hubs on unfair terms.

Figure 1 Structure of the CPPI



Source: World Bank.

Several exclusion criteria were applied to the port call data. Call size groups with less than 0.5 percent of all data per given ship size group were excluded from the CPPI calculations due to insufficient port calls. To be included, port calls had to have a reasonable berth productivity – with berth moves per hour between 10 and 650. This threshold is a data-quality filter, intended to remove erroneous or miscoded observations. In addition, ports with fewer than 24 container calls per year in the dataset are excluded from the calculations.

4.2 *The Administrative Approach*

The administrative approach is a direct measure of port efficiency. For each port, average hours in port are computed within each call size group and compared against the global average for that group, for a given ship size. Results across call size groups are weighted by the share of port calls in each group, and results across ship size categories are aggregated using the Fuel Consumption Index as the weighting scheme.

Step 1: Define port time

Port time includes all time from a vessel's AIS-detected arrival at the port limit or anchorage zone to the time it leaves the berth (arrival to departure).

These six time stamps are aggregated into:

- Arrival hours (time elapsed between key events 1 and 3), which consist of waiting time (if applicable), and steam-in time
- Berth hours (time elapsed between key events 3 and 5), which consist of cargo operation time and idle time at berth without cargo loading
- Departure hours (time elapsed between key events 5 and 6), which consist of time spent in the port after leaving the berth (while this period is recorded, it is not included in the CPPI calculations)

The administrative approach uses aggregated arrival and berth hours. To handle missing values, the administrative approach imputes missing values using mean values within the same ship size group and port for arrival hours, and the same ship size and call size group for berth hours.

Step 2: Mean-center by ship size and call size

To ensure fair comparisons, the raw port time is adjusted using two main operational controls:

- Ship size: categorized into five predefined groups by TEU capacity
- Call size: categorized into ten predefined groups by number of container moves per port call (load + discharge + restow)

Within each call size group, the port's average port hours are compared with the group's average port hours as a negative or positive quantity of hours. The result of that comparison is weighted by the ratio of port calls in each call size group for the entire group of ports. This is then summed, and the results are port-level scores per ship size category.

Step 3: Aggregate port-level scores

As performance may vary significantly by vessel size, separate indices are calculated per ship size group. The final CPPI score is a weighted average of the port's performance across all relevant ship size groups.

The ship size groups are weighted using a Fuel Consumption Index (see Table 1) to differentiate the importance and significance of improved performance on larger ships compared to smaller ones, based on the relative fuel consumption between different ship sizes. This reflects the greater operational and economic significance of reducing port time for larger ships relative to smaller ones.

For each ship size group, a typical midrange example ship was selected. Based on the expected deployment of such ships, the index defines and weights a range of sea legs, using a typical pro forma service speed, and considers the impact on fuel consumption that one hour longer (or shorter) in port would likely yield. The index weight then suggests that, for example, it is 2.57 times more costly to recover an additional hour of port time at sea for a ship with a capacity of over 13,500 TEUs than it would be for a ship in the 1,501–5,000 TEU capacity range.

Table 1 Fuel Consumption Index

NOMINAL TEU CAPACITY RANGE	EXPECTED DEPLOYMENT	SEA LEG	WEIGHT (PERCENT)	INDEX WEIGHT
Less than 1,500 TEUs	Feeders Intra-regional	Singapore–Surabaya	25	0.46
		Rotterdam–Dublin	25	
		Kingston–Port-au-Prince	25	
		Busan–Qingdao	25	
1,500 to 5,000 TEUs	Intra-regional Africa Latin America Oceania Transatlantic	Shanghai–Manila	30	1.00
		Rotterdam–Genoa	30	
		Algeciras–Tema	10	
		Charleston–Santos	10	
		Xiamen–Brisbane	10	
		Felixstowe–New York	10	
5,000 to 8,500 TEUs	Africa Latin America Oceania Transatlantic Asia–Middle East	Hong Kong–Tema	20	1.54
		Charleston–Santos	20	
		Xiamen–Brisbane	20	
		Felixstowe–New York	20	
		Shanghai–Dubai	20	
8,500 to 13,500 TEUs	Transpacific Asia–Middle East Asia–Mediterranean	Busan–Charleston (via Panama)	25	1.97
		Hong Kong–Los Angeles	25	
		Shanghai–Dubai	25	
		Singapore–Piraeus	25	
Greater than 13,500 TEUs	Asia–Mediterranean Asia–North Europe Transpacific	Singapore–Piraeus	40	2.57
		Singapore–Rotterdam	40	
		Hong Kong–Los Angeles	20	

Source: World Bank, based on data provided by S&P Global Market Intelligence.

4.3 The Statistical Approach

The statistical approach applies non-negative matrix factorization (NMF), a widely used multivariate data reduction technique, to identify underlying (“latent”) dimensions of port performance that jointly explain the observed data. Instead of comparing each port to a

benchmark or group average, the method analyses all ports and all call size categories simultaneously, allowing common efficiency patterns to emerge from the data itself. This helps capture structural relationships – such as systematic differences between small and large vessel calls – that are not visible when indicators are assessed one by one.

The starting point is the same as in the administrative approach: for each port, the average arrival time and average berth time are calculated for each ship size and call size category (see Figure 1). Because not all ports receive calls in every category, some values are missing. These gaps are filled using an Expectation–Maximization (EM) algorithm, which estimates plausible values based on observed patterns in the dataset while preserving internal consistency. A non-negativity constraint is imposed throughout the process so that all estimated times remain operationally meaningful (i.e. non-negative) and comparable with the administrative results.

Once the dataset is complete, arrival and berth times are added to obtain total port time. For each ship size group, the port times are then mean-centered and weighted in the same way as under the administrative approach, ensuring consistency between the two methodologies. This produces a matrix X in which each row represents a port and each column represents a call size category, with entries corresponding to weighted port times.

NMF then decomposes this matrix into two smaller non-negative matrices: one capturing a limited number of latent performance factors, and the other capturing how strongly each port is associated with each factor. Intuitively, these latent factors can be thought of as common “profiles” of port performance (for example, consistently fast turnaround across all call sizes, or relatively better performance for large vessels than for small ones). Each port’s performance is thus expressed as a combination of these underlying patterns.

A simple numerical illustration helps clarify the logic. Suppose three call size categories are observed for a given ship size, and a port has weighted port times of 10, 20, and 30 hours respectively. Rather than treating these three numbers independently, NMF may identify two latent factors, for example, a “general efficiency” factor and a “large-call penalty” factor. The port’s observed values are then approximated as a combination of these two factors, with non-negative weights. Summing the resulting factor contributions yields a single index for that ship size category. In practice, this process is carried out across all ports simultaneously. The derived factors reflect global patterns rather than individual cases.

As in the administrative approach, a port-specific index is first obtained for each ship size group. The overall statistical CPPI score is then computed as a weighted average of these indices across ship sizes, using the same weights as before. This ensures that differences between the two approaches arise only from the method of aggregation, not from differences in underlying data or weighting schemes.

The approach follows the methodology set out in the CPPI analytical framework (see Ashley et al., 2023) and builds on established techniques in multivariate statistics (Bartholomew et al., 2011) and non-negative matrix factorization (Lee and Seung, 1999, and Gillis, 2020). While more complex than the administrative method, it provides an internally consistent use of the available data, particularly when performance indicators are interrelated.

4.4 Combining the Two Indices into the CPPI

Both indices are highly correlated (see Figure 2 and Figure 3). The reported Container Port Performance Index CPPI is the arithmetic average of the two independent approaches – the administrative and the statistical. Ports are ranked in descending order of their combined CPPI score. This final ranking constitutes the overall CPPI ranking reported in the index report.

By averaging the two indices before ranking, the CPPI benefits from the complementary strengths of each approach: the transparency and interpretability of the administrative method, and the statistical robustness of the matrix factorization method. Presenting the two component indices alongside their average for transparency also allows readers to assess the degree of consistency between methodologies for any given port.

Figure 2 Correlation between the Statistical Index and the Administrative Index, 2025

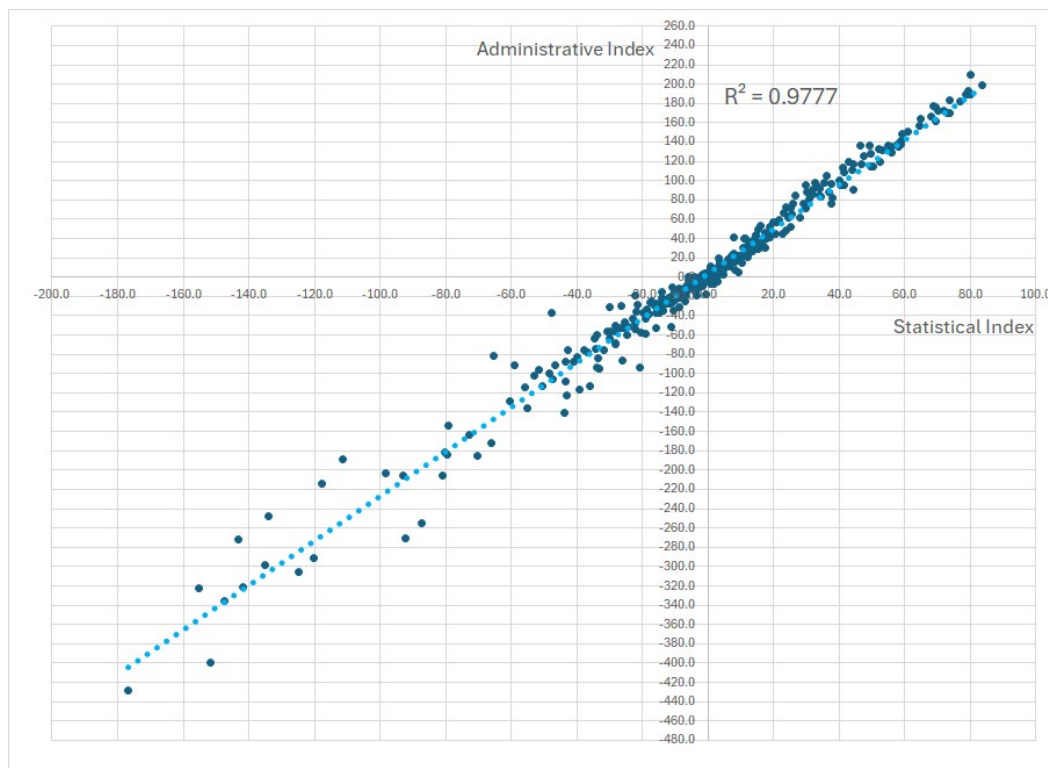
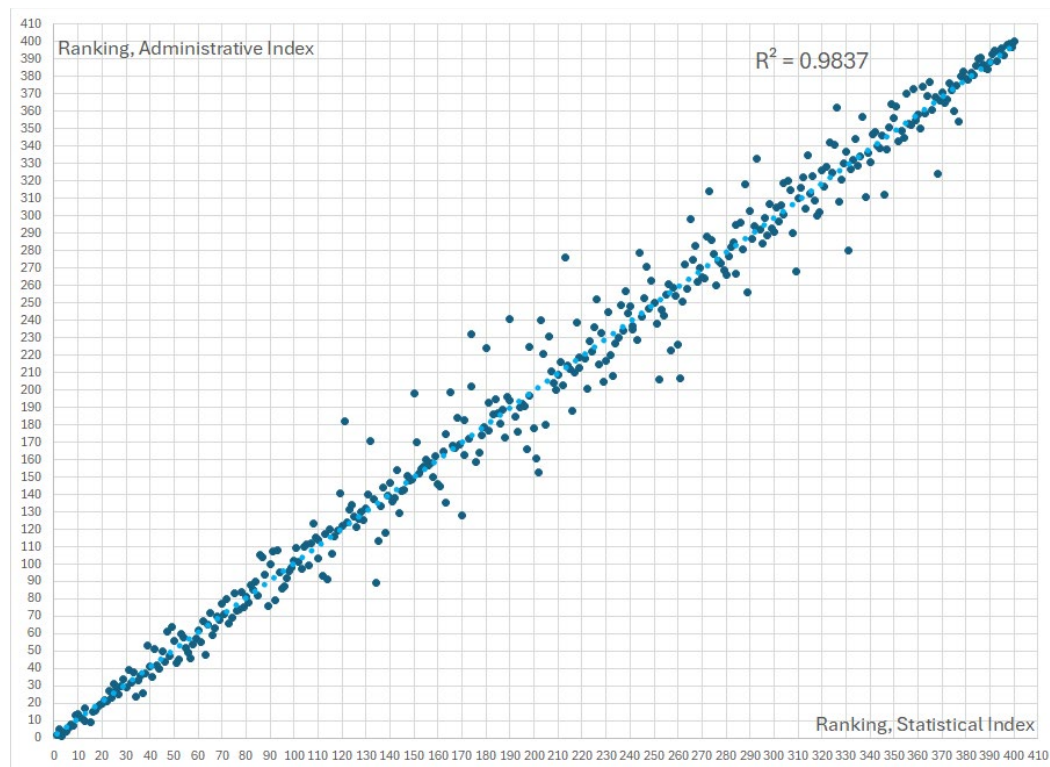


Figure 3 Correlation between the rankings under the Statistical Index and the Administrative Index, 2025



5. Multi-Year Time Series and Standardization

From the 2024 edition onwards, the CPPI includes a standardized multi-year time series covering the full period starting from 2020. A fixed reference year allows stakeholders to assess whether a port's performance has genuinely improved or deteriorated over time, rather than simply shifting relative to other ports in a given year's sample.

In earlier editions, the CPPI scores were internally normalized to a mean of zero within each individual year, making year-to-year comparisons across editions unreliable – a score improvement could reflect genuine operational progress or simply a change in the composition of the port sample.

To address this, a single reference distribution is used to anchor the time series. Specifically, the combined CPPI is standardized so that the average combined CPPI across all ports equals zero in the chosen reference year, with all other years expressed relative to this baseline. In practice, this means that the time spent in port in each call size group is based on the reference year average time spent in port. The approach is applied in each ship size group, for both the administrative and the statistical approaches. This enables consistent trend analysis across the full time series.

This means that a positive CPPI value indicates above-average performance relative to the reference year's distribution, while a negative value indicates below-average performance. Changes in a port's CPPI score over time reflect genuine shifts in performance relative to the fixed baseline, not changes in peer composition.

The reference year for the current standardization is 2024. Should the reference year be updated in a future edition, historical scores would be rebased accordingly.

6. Interpretation and Limitations

The CPPI is designed as a benchmarking tool. It identifies performance gaps and helps prioritize areas for further investigation, but it does not, by itself, explain the causes of those gaps. Extended port times may reflect factors that are within a port's control, such as productivity, equipment availability, or operational coordination. However, longer port times may also reflect factors that are largely exogenous, such as vessel arrival patterns, tidal constraints, or labor regulations.

Users are encouraged to treat the CPPI as a starting point for deeper analysis rather than a definitive performance verdict. Regional and throughput-based sub-rankings provide additional context for interpreting results within comparable peer groups.

7. References

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